

ACOUSTIC SCENE CLASSIFICATION: AN EVALUATION OF AN EXTREMELY COMPACT FEATURE REPRESENTATION

*Gustavo Sena Mafra**

Universidade Federal de Santa Catarina
Florianópolis, Santa Catarina, Brazil
gsenamafra@gmail.com

Ngoc Q. K. Duong[†], Alexey Ozerov, Patrick Pérez

Technicolor
975 Avenue des Champs Blancs CS 17616
35576 Cesson-Sévigné, France
firstname.lastname@technicolor.com

ABSTRACT

This paper investigates several approaches to address the acoustic scene classification (ASC) task. We start from low-level feature representation for segmented audio frames and investigate different time granularity for feature aggregation. We study the use of support vector machine (SVM), as a well-known classifier, together with two popular neural network (NN) architectures, namely multilayer perceptron (MLP) and convolutional neural network (CNN). We evaluate the performance of these approaches on benchmark datasets provided from the 2013 and 2016 Detection and Classification of Acoustic Scenes and Events (DCASE) challenges. We observe that a simple approach exploiting averaged Mel-log-spectrograms and SVM can obtain even better results than NN-based approaches and comparable performance with the best systems in the DCASE 2013 challenge.

Index Terms— Acoustic scene classification, Audio features, Multilayer Perceptron, Convolutional Neural Network, Support Vector Machine.

1. INTRODUCTION

Acoustic scene classification (ASC), a particular form of audio classification, consists in using acoustic information (audio signals) to infer the context of the recorded environment [1]. Examples of such environments are bus, office, street, etc... It offers a wide range of applications in connected home, *e.g.* expensive video cameras can be replaced by cheap microphones for monitoring daily activity, and for smartphones, *e.g.* they could automatically switch to silence mode during a meeting or automatically increase the sound volume in a noisy environment. However, real-life ASC is not a trivial task as recognising a greater variety of sounds in both indoor and outdoor environments would require a new set of strategies and adjustments of existing machine learning techniques to make the most out of the available data.

While speaker identification [2], speech recognition [3], and some audio classification tasks in music information retrieval such as music genre recognition [4, 5] or music instrument recognition [5] have been studied for a long time, the real-life ASC task has become active quite recently in the research community. While the classification task itself has been studied since at least 2002 [6],

it was only recently that efforts were made to provide a benchmark for the task, with the new initiative of the DCASE challenges in 2013 and 2016. Various techniques have been proposed to tackle the problem with the use of different acoustic features (*e.g.* cochleogram representation, wavelets, auditory-motivated representation, features learned by neural networks) and different classifiers (*e.g.* Support Vector Machine (SVM), Gaussian Mixture Model (GMM), Hidden Markov Model (HMM)) [7]. One of the most popular approaches, known as bag-of-frames (BOF) approach [8, 9] is used as a baseline in the DCASE challenge, and exploits the long-term statistical distribution (by GMM) of the short-term MFCCs.

Besides the DCASE challenge, nonnegative matrix factorization (NMF) was recently exploited for sound event detection in real life recordings [10]; recurrent neural networks (RNN) were investigated for polyphonic sound event detection in real life recordings [11]; and deep neural networks (DNN) have been developed for sensing acoustic environment [12]. It would be interesting to note that while DNNs [13, 14] were recently applied with great success to many different audio, visual and multimedia tasks, it was less investigated within the DCASE 2013 challenge and one of the reasons would be the lack of a substantial amount of labeled data for training.

This paper aims to study the use of well-established low-level acoustic feature representations and different machine learning techniques, including DNN-based methods and SVM, for the ASC task. While most existing approaches extract an acoustic feature vector for each short-term audio frame, then perform a frame-based classification based either on BOF over GMMs [8, 9] or simple majority voting [15, 16, 7], we investigate the use of an another feature representation, *i.e.* a single vector for a whole audio scene, aiming an extremely compact representation that greatly reduces the computational cost for the whole ASC system, since the number of examples to be used to train the classifier is drastically reduced. We evaluate the use of this compact feature with SVM and MLP on both DCASE 2013 and DCASE 2016 datasets and the performance are more or less equivalent to a frame-based approach with majority voting strategy. Furthermore, it results in classification accuracy comparable to the best systems participating in the DCASE 2013 challenge.

The rest of the paper is organized as follows. In Section 2 we present the general framework which involves different approaches for feature extraction and classification. Experiment results on DCASE dataset obtained by our approaches and some state-of-the-art methods are discussed in Section 3. We finally conclude in Section 4.

*Part of this work has been done while the first author was with Technicolor.

[†]Email: quang-khanh-ngoc.duong@technicolor.com

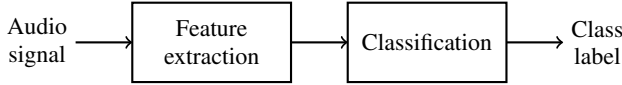


Figure 1: General workflow of the acoustic scene classification framework.

2. ACOUSTIC SCENE CLASSIFICATION FRAMEWORK

The general workflow of an ASC system is usually divided into two major steps as shown in Fig. 1. In the first, the feature extraction step, various types of hand-crafted representations have been considered in the literature such as chroma, pitch, spectrograms, zero-crossing rate, and linear predictive coding coefficients [1]. Among them, features based on Mel-frequency Cepstrum Coefficients (MFCCs) computed for each short-time frame are arguably the most common one, as it can be seen by the DCASE 2013 Challenge, where out of 12 systems submitted, at least 7 involved MFCCs [1]. More recent DNN-based approaches usually attempt to learn higher level features from these low-level signal representations [12, 17]. In the classification step, popular classifiers include SVM and GMM [7]. In the following, we will first describe the standard Mel-log-spectrogram, as the low-level feature used in this work, and the proposed compact representation from it in Section 2.1. We then briefly present some exploited classification approaches in Section 2.2. The choice of hyperparameters for both feature extraction and classifiers is discussed in Section 2.3.

2.1. Feature extraction

The time domain audio signal $x(n)$ is first transformed into the frequency domain by means of the short-term Fourier transform (STFT) as

$$\text{STFT}\{x\}(m, \omega) = \sum_{n=-\infty}^{+\infty} x(n)w(n - mL)e^{-j\omega n} \quad (1)$$

where $w(n)$ is a window function (which is Hanning window in our implementation), m denotes frame index and L the frame shift. The spectrogram is then defined as

$$\mathbf{S}(m, \omega) = |\text{STFT}\{x\}(m, \omega)|^2 \quad (2)$$

In our DNN-based system, we use spectrogram with a logarithmic amplitude scale (named log-spectrogram) as the frame input feature which is computed as

$$\mathbf{F}_{\text{Log-spec}}(m, \omega) = \log(\mathbf{S}(m, \omega)). \quad (3)$$

In our other systems, we first map the spectrogram $\mathbf{S}(m, \omega)$ into the auditory-motivated Mel frequency scale - denoted by $\mathbf{MS}(m, \omega)$, then transform it into logarithmic scale as

$$\mathbf{F}_{\text{Mel-log-spec}}(m, \omega) = \log(\mathbf{MS}(m, \omega)). \quad (4)$$

Note that with the CNN-based system, we use the raw log-spectrogram as the input feature in order to give flexibility for the CNN to learn a higher level feature representation optimized for the ASC task. For SVM-based systems we have tested four different features: spectrogram, log-spectrogram, Mel-log-spectrogram, and MFCC, and found that the two last ones result in a very similar

ASC performance that outperforms the two first ones. As the Mel-log-spectrogram is simpler to compute than the MFCC, we use it as the main acoustic feature in this paper. Finally, we propose to average the feature vectors for all frames so as to present a whole audio example by an extremely compact feature vector whose entries are computed as

$$\mathbf{f}_{\text{Avg-mel-log-spec}}(\omega) = \frac{1}{M} \sum_{m=1}^M \mathbf{F}_{\text{Mel-log-spec}}(m, \omega). \quad (5)$$

This type of averaging of features is very straight-forward and has already been used in past works [18][19] in ASC. However, no submitted systems in the DCASE 2013 Challenge made use of it, misleadingly pointing to a lack of efficiency of this method.

2.2. Classification approaches

2.2.1. Support vector machine

SVM has been known as one of the most popular classifiers for many different tasks. It was also widely used in the DCASE 2013 challenge [7]. In our work, we used SVM as a benchmark classifier to evaluate the effectiveness of different features, as mentioned in Section 2.1, as well as to obtain the optimal choice of hyperparameters (*e.g.* the window size and the number of Mel-frequency coefficients) for the considered task.

In our implementation, we train SVMs using a coordinate descent algorithm [20] and following a one-vs-the-rest scheme to perform classification of multiple classes [21]. We have tested SVM with linear kernel and Gaussian radial basis function (RBF) kernel and found that the linear kernel works slightly better than RBF kernel for the DCASE 2013 dataset.

2.2.2. Multilayer Perceptron

Multilayer Perceptron (MLP) is a fully connected feedforward artificial neural network architecture that maps sets of input data onto a set of appropriate outputs. It can be seen as a logistic regression classifier where the input is first transformed using a non-linear transformation [22, 23]. A typical set of equations for an MLP is the following. Layer k computes an output vector $\mathbf{h}^k$ using the output $\mathbf{h}^{k-1}$ of the previous layer, starting with the input $\mathbf{x} = \mathbf{h}^0$,

$$\mathbf{h}^k = f(\mathbf{W}^k \mathbf{h}^{k-1} + \mathbf{b}^k) \quad (6)$$

where $\mathbf{b}^k$ denotes a vector of offsets (or biases) and $\mathbf{W}^k$ a matrix of weights. The function f is called the activation function and it is applied element-wise. Common options for it are sigmoid function, hyperbolic tangent, and rectified linear unit (ReLU). The latter, *i.e.* $f(x) = \max(0, x)$, was used to obtain the results reported in this document.

The top layer output is used for making a prediction and is combined with the groundtruth label into a loss function. We use softmax as the classification layer and the log-likelihood loss function regularized with ℓ_1 and ℓ_2 penalties. This cost function is then optimized using mini-batch stochastic gradient descent (SGD) with an adaptive learning rate [24] and dropout is performed between the hidden layers [25].

2.2.3. Convolutional Neural Network

Convolutional Neural Network (CNN) is a type of neural network designed to exploit the redundancy and correlation between neighbour units. It has gained great success in different fields such as image and video recognition, natural language processing, speech recognition, etc., [14]. This motivates us to investigate the use of CNN for the ASC task in this work.

We trained CNNs over the log-spectrogram of the signals with a structure of vertical filters, *i.e.* the frequency bins can also be interpreted as a CNN channel (as in RGB channels for images) instead of a dimension and the convolution is ran over the time axis. This type of structure was proposed for music recommendation in Spotify¹ and is justifiable by the fact that an audio “pattern” detected in a high-frequency region is usually different from that same pattern in a low-frequency region. Thus it is desirable to model the vertical filters to extract more meaningful information from the spectral representation. More details about the implemented CNN architecture can be found in Section 3.1.

2.3. Hyperparameter optimization

The choice of hyperparameters in each step of the ASC system or in any machine learning task can significantly affect the final classification result. Such hyperparameters are *e.g.* the window length and hop length in the STFT computation for feature extraction, the regularization parameter for SVM, the number of hidden units in an MLP, and the step size for the SGD algorithm in DNN based methods. The conventional strategy of tuning these parameters manually would not be feasible as it requires a great number of trials so that all parameters can be optimized together. Thus, in this work we incorporate a Bayesian optimization [27] method to find these parameters altogether. The algorithm models the generalization accuracy of a classifier as a function of the corresponding parameters, and finds the optimal parameters that maximize the expected accuracy given the observed dataset.

In our implementation, we use Hyperopt [28], a Python library for optimizing hyperparameters in machine learning algorithms, with the Tree of Parzen Estimators (TPE) [29], an algorithm that falls into the class of sequential model-based optimization (SMBO) [30] algorithms. The TPE algorithm performs cross-validation with the development datasets of DCASE 2013 and DCASE 2016 and finds an optimal set of hyperparameter values. It is interesting to note that the optimal window size for STFT computation found by the TPE algorithm is quite long, *i.e.* about half of a second. This can be explained by the fact that the acoustic events are more spread in time compared to *e.g.* speech which is very localized so as the window length used for STFT is usually much smaller.

3. EXPERIMENTS

We evaluate the ASC performance of our four implementing systems with the benchmark DCASE 2013 dataset, which allows to compare with the state-of-the-art approaches participating in the challenge, in Section 3.1. We then present the result with DCASE 2016 dataset in Section 3.2. Our first system (named *Proposed SVM-A*) uses an extremely compact feature as the Mel-log-spectrogram coefficients averaged for all frames, and SVM with a linear kernel as classifier. The second system (named *Proposed SVM-V*) performs frame classification by SVM with a linear kernel,

then majority voting in the end. The third system (named *Proposed MLP*) takes the compact averaged Mel-log-spectrogram as input, learns an intermediate feature representation by MLP, then classifies by softmax as the last layer of the MLP. The fourth system (named *Proposed CNN*) takes log-spectrogram as low-level input feature, learns higher feature representations by CNN layers, then classifies by softmax.

3.1. Results with the DCASE 2013 dataset

The DCASE 2013 dataset consists of 30-second audio segments belonging to 10 classes. Each class has 10 segments in the development set and 10 other examples in the test set [31].

The ASC performance was evaluated in terms of the classification accuracy, averaged over all classes, and shown in Table 2. Note that as we did not have access to the groundtruth labels of the test set at the time of these experiments, we evaluated our systems averaging with the standard 5-fold cross-validation on the development set only, while results for most other approaches in the table are obtained with the test set [7]. Some hyperparameters for each systems were found by the Bayesian optimization method presented in Section 2.3. More detailed settings for each system are as follows. The window length for the STFT was set by 0.57 seconds and 0.41 seconds for the SVM-A and SVM-V system, respectively, the number of Mel-frequency coefficients is about 1900, the regularization parameter C in SVM for SVM-A and SVM-V were 0.98 and 0.62, respectively. MLP had one hidden layer with 677 units, dropout rate and learning rate for parameter training was set by 0.08 and 0.011, respectively. CNN had 3 convolutional layers, the number of filters for each layer are 50, 29, and 19, respectively, and the max-pooling ratios between layers are 3, 4, and 3.

As it can be seen, the two systems based on SVMs outperform the ones based on NNs. This can be explained by the fact that the dataset may be not large enough for training DNNs directly. Three of our proposed systems (SVM-A, SVM-V, and MLP) achieve comparable performance with some of the best performing approaches in the DCASE 2013 Challenge - as we suppose that there is not much difference between development set and the test set. Moreover, we achieve higher accuracy than Li *et al.* [16] in the same development set. Finally, we note that the proposed feature, which is extremely compact so as to represent a whole 30-second audio segment by just a single vector, can be sufficient for the classification as the SVM-A and MLP obtained 75% and 72% accuracy, respectively.

3.2. Results with the DCASE 2016 dataset

The DCASE 2016 dataset is structured in a similar way as the DCASE 2013 dataset. However the number of acoustic classes is extended to 15, and the number of examples for each class is significantly enlarged to 78 for the development set and 26 for the test set.

The results for development set obtained by our four systems are shown in Table 2, where the best performance of 80% is achieved by the SVM-A system with a window length of 0.42 seconds and a hop size of 0.14 seconds for the STFT computation. This result confirms again the benefit of using the proposed compact feature representation and the use of a long window for the spectral transformation. The MLP, which obtains similar performance as the baseline, had two hidden layers with 66 and 199 units, respectively, SGD was used for parameter training with learning rate of

¹<http://benanne.github.io/2014/08/05/spotify-cnns.html>

Method	Acoustic feature	Classifier	Accuracy
Baseline	MFCC	"bag-of-frames" GMM	55%
Geiger <i>et al.</i> [15]	Diverse features	SVM + majority voting	69%
Roma <i>et al.</i> [26]	MFCC with Recurrence Quantification Analysis	SVM	76%
Proposed SVM-A	Averaged Mel-log-spectrogram	Linear SVM	75%
Proposed SVM-V	Frame Mel-log-spectrogram	Linear SVM + majority voting	78%
Proposed MLP	Averaged Mel-log-spectrogram	MLP with softmax as classification layer	72%
Proposed CNN	Log-spectrogram	CNN with softmax as classification layer	62%

Table 1: Acoustic scene classification results with DCASE 2013 test dataset (for state-of-the-art approaches) and development dataset (for our proposed approaches). Note that other submitting systems resulting in less classification accuracy are not mentioned in the table.

Method	Acoustic feature	Classifier	Accuracy
Baseline	MFCC	"bag-of-frames" GMM	75%
Proposed SVM-A	Averaged Mel-log-spectrogram	Linear SVM	80%
Proposed SVM-V	Frame Mel-log-spectrogram	Linear SVM + majority voting	78%
Proposed MLP	Averaged Mel-log-spectrogram	MLP with softmax as classification layer	75%
Proposed CNN	Log-spectrogram	CNN with softmax as classification layer	59%

Table 2: Acoustic scene classification results with DCASE 2016 development dataset.

	Beach	Bus	Cafe/restaurant	Car	City center	Forest path	Grocery store	Home	Library	Metro station	Office	Park	Residential area	Train	Tram
Beach	47	0	3	4	5	4	1	0	1	0	3	4	2	0	2
Bus	2	61	0	4	1	0	0	0	0	2	0	2	2	3	1
Cafe/restaurant	4	0	46	0	0	2	19	4	0	0	0	2	0	0	1
Car	0	2	0	71	0	0	0	0	0	0	0	0	1	0	4
City center	0	0	0	0	75	0	0	0	0	2	0	0	1	0	0
Forest path	1	0	0	0	0	75	0	0	0	0	0	0	2	0	0
Grocery store	0	0	0	0	2	0	76	0	0	0	0	0	0	0	0
Home	6	0	1	0	0	13	1	30	7	0	3	6	9	1	1
Library	0	1	0	0	0	0	0	0	72	1	0	0	0	4	0
Metro station	0	0	0	0	0	0	0	0	0	76	0	2	0	0	0
Office	0	0	0	0	0	0	0	8	1	0	69	0	0	0	0
Park	9	0	5	0	2	1	0	1	0	0	0	49	11	0	0
Residential area	5	0	0	0	9	7	0	0	2	0	0	19	35	0	1
Train	17	3	0	0	2	0	0	0	1	0	0	1	0	49	5
Tram	2	0	0	0	0	0	0	0	0	2	0	0	0	0	74

Figure 2: Confusion matrix of the SVM-A method in the development set of the DCASE 2016 database after a 4-fold cross-validation over 78 samples of each class.

0.003 and batch size of 100, weights for ℓ_1 and ℓ_2 penalties were 10^{-5} and 10^{-4} , respectively. The CNN, with the same configuration used for the DCASE 2013 dataset, still resulted in the lowest performance. These four systems will also be tested with the test set for participating in the DCASE 2016 challenge.

The confusion matrix for SVM-A is shown in Fig. 2, where rows are groundtruth, columns are the inferred class label, and values are number of the classified acoustic scene. As it can be seen, some environments containing a specific type of noise (such as car, metro station, forest path) are quite easy to recognize, while some

others (such as home, residential area, park) are quite confusing.

4. CONCLUSION

In this article we present several approaches for the ASC task, targeting on fast systems working with very compact feature representations so that ASC can be implemented *e.g.* in smartphones. We investigate the use of Bayesian optimization for hyperparameter optimization and find its benefit in *e.g.* choosing the optimal window length for STFT or setting DNN parameters. By evaluating on benchmark DCASE datasets, we find that (1) a long window size for spectral transformation is more relevant for the environmental acoustic scenes, (2) a very compact feature representation by long-term temporal averaging of Mel-log-spectrogram coefficients would be sufficient for the task compared to more complicated approaches, and (3) similar accuracies are found for the two datasets, possibly owing to the similarities between these datasets or to the robustness of the proposed systems. Finally, it is worth noting that DNN approaches have not reached the same performance of the more classical SVM based systems so far. Thus future work would be devoted to investigate transfer learning strategies for DNN based systems where part of the DNN can be initially learned by a large amount of external audio data.

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